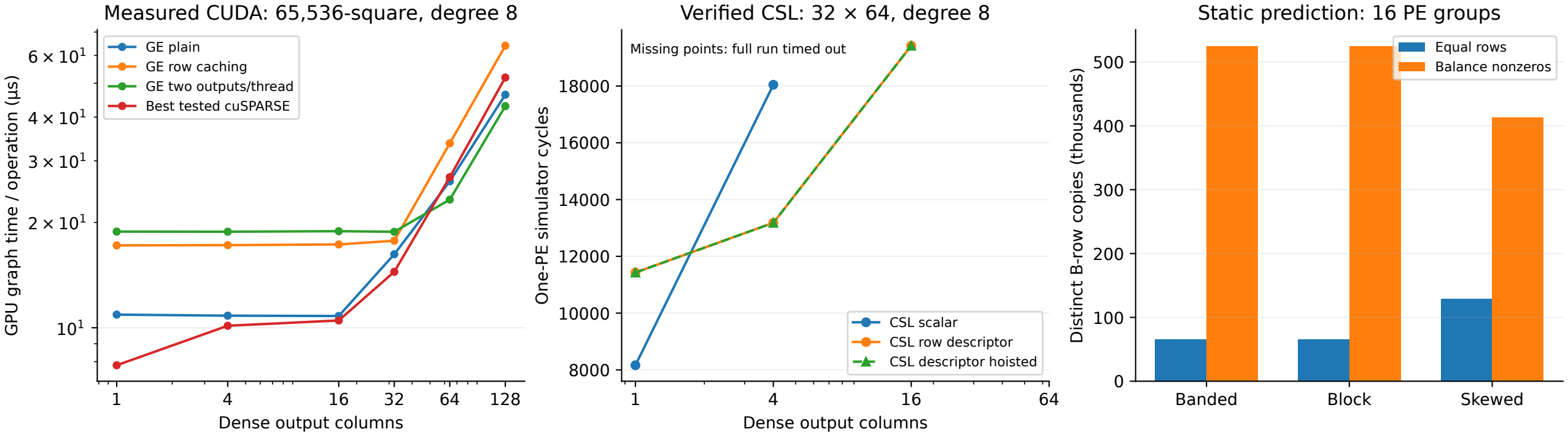
Measured CUDA: 65,536-square, degree 8
GE plain
GE row caching
GE two outputs/thread
Best tested cuSPARSE
GPU graph time / operation (µs)
6 × 10^1
4 × 10^1
3 × 10^1
2 × 10^1
10^1
1
4
16
32
64
128
Dense output columns
Verified CSL: 32 × 64, degree 8
Missing points: full run timed out
One-PE simulator cycles
18000
16000
14000
12000
10000
8000
CSL scalar
CSL row descriptor
CSL descriptor hoisted
1
4
16
64
Dense output columns
Static prediction: 16 PE groups
Equal rows
Balance nonzeros
Distinct B-row copies (thousands)
500
400
300
200
100
0
Banded
Block
Skewed